

NETWORK TOPOLOGIES — ASSIGNMENT

Comprehensive Analysis, Comparisons, and Practical Scenarios

Task 1: Topology Diagrams and Real-World Applications

- **Star Topology**
 - **Diagram Structure:** All client nodes are connected independently to a central hub, switch, or router. If viewed geometrically, it resembles spokes radiating from a central wheel axle.
 - **Real-World Application / Service: WhatsApp (Messaging and Media Delivery via Centralized Cloud Servers)**
 - **Explanation:** WhatsApp relies on a star-like client-server architecture where each user device connects directly to centralized WhatsApp servers. This ensures fast message relaying, robust authentication, and reliable delivery even if individual recipient devices go offline temporarily.
- **Bus Topology**
 - **Diagram Structure:** All devices are connected to a single central communication line, known as the 'backbone' cable or bus, with terminators placed at both ends to prevent signal bounce.
 - **Real-World Application / Service: Ethernet Local Area Network (Legacy Office Networks / Simple Sensor Arrays)**
 - **Explanation:** A bus topology can be utilized in simple, legacy industrial sensor networks or small-scale classroom setups where data packets travel sequentially along a single shared coaxial cable. It minimizes initial wiring requirements when connecting devices along a linear path.
- **Mesh Topology**
 - **Diagram Structure:** Every node has a dedicated point-to-point physical or logical link connecting it to multiple other nodes or all other nodes in the network, ensuring multiple redundant pathways.
 - **Real-World Application / Service: IPL Live Streaming (Content Delivery Network - CDN Edge Routing)**
 - **Explanation:** While end-user video streaming operates on client-server models, the underlying CDN backbone transmitting IPL live feeds across global edge servers uses full mesh or hybrid mesh interconnections. This guarantees zero single points of failure and ultra-low latency failover if one regional peering link becomes congested.

Task 2: Comprehensive Topology Comparison Table

The following table provides an in-depth comparison of Star, Bus, Ring, and Mesh topologies across four pros, four cons, best use cases, and real-world app examples.

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case	Real-World App Example

Star	1. Easy to install and wire. 2. Failure of one node does not affect others. 3. Simple to troubleshoot. 4. High performance with dedicated links.	1. High dependency on central hub/switch. 2. Central device failure disables entire network. 3. Requires more cable than Bus. 4. Cost of hub/switch adds to setup.	Local Area Networks (LANs) in offices, homes, and enterprises requiring reliable centralized control.	Instagram (Client connects to Instagram's central media/feed servers via Wi-Fi/cellular router)
Bus	1. Uses very little cabling. 2. Easy to set up in small linear spaces. 3. Cost-effective for small networks. 4. Easy to understand and expand linearly.	1. Cable break brings down entire network. 2. Difficult troubleshooting. 3. Performance degrades with heavy traffic. 4. Security risk (all nodes see broadcast traffic).	Small, temporary, or legacy networks with low traffic and budget constraints.	Zomato (Internal legacy restaurant kitchen display terminal daisy-chains)
Ring	1. Equal access for all devices (no collisions). 2. Performs better than bus under heavy load. 3. Easy to identify broken cable segments. 4. No central server required.	1. Single device/cable break halts entire ring. 2. Adding/removing nodes disrupts network. 3. Data latency increases with more nodes. 4. Difficult to troubleshoot configuration.	Token-ring industrial control systems and closed campus loops.	Paytm (Closed-loop smart card transit gate turnstiles communicating in a serial loop)
Mesh	1. Maximum fault tolerance and redundancy. 2. No single point of failure. 3. High privacy and security. 4. Handles massive data traffic simultaneously.	1. Extremely expensive to install and wire. 2. Complex installation and configuration. 3. High maintenance overhead. 4. Redundant links increase hardware cost.	Critical backbone networks, WANs, military comms, and high-availability data centers.	WhatsApp (Peer-to-peer media transfers and decentralized node routing in distributed datacenters)

Task 3: Multiplayer Mobile Game Server Architecture Design

Chosen Topology: Hybrid Star-Mesh Topology (Client-Server Star for matches combined with a Mesh backbone across regional data centers).

Justification & Impact Analysis:

- **Speed (Latency):** Connecting players in a match to a centralized regional game server (Star topology) ensures minimal ping and deterministic state synchronization. If a full mesh were used between 100 players in a Battle Royale game, bandwidth and packet processing overhead on mobile devices would cause extreme lag.
- **Reliability:** Using a mesh backbone interconnecting regional cloud datacenters guarantees high reliability; if one server region fails, player traffic is instantly rerouted to an adjacent node without dropping the session.
- **Cost:** While maintaining distributed cloud servers incurs recurring infrastructure expenses, it is vastly more cost-effective and scalable than building dedicated peer-to-peer mesh circuits for millions of concurrent mobile users.

Task 4: Flipkart-Style Warehouse Network Architecture

Scenario Recommendation: Star Topology using a Central Gigabit Switch.

Diagram Structure (Text Representation):

```
[ Terminal 1 ] [ Terminal 2 ] [ Terminal 3 ] [ Terminal 4 ]
      \       |       |       /
      ----- [ CENTRAL SWITCH ] -----
      /       |       |       \
[ Terminal 5 ] [ Terminal 6 ] [ Terminal 7 ] [ Terminal 8 ]
```

Reasoning & Analysis:

1. **Cost-Effective:** Requires affordable Cat6 Ethernet cables and a standard 8-to-16 port network switch.
2. **Ease of Troubleshooting:** If a barcode scanner or order processing terminal fails, only that specific cable or workstation is affected; the remaining 7 computers continue operating without interruption.
3. **Future Expansion:** Adding more packing stations or inventory terminals in the future is seamless—simply plug a new cable into an available port on the central switch.